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Edgecore SONiC
[Edgecore SONiC] Installation & Upgrade image 

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Edgecore SONiC Download
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Open Networking Switch
How to install SONiC images, how to upgrade, how to set start-up image when you have multiple images in your device and how to remove the image you don't need.
Open Networking Switch Download (ONIE, BIOS, CPLD, FPGA, BMC, ACCTON-DIAG)

- Install SONiC from HTTP or TFTP via ONIE
- Install SONiC from USB via ONIE
- Edgecore SONiC Upgrade
- Edgecore SONiC Upgrade via USB
- Specified the startup image
- Remove installed image

Edgecore and IgniteNet Wireless

Tested model & firmware version:

- Switch model name:

AS7326-56X

- Edgecore SONiC version:

202006.4

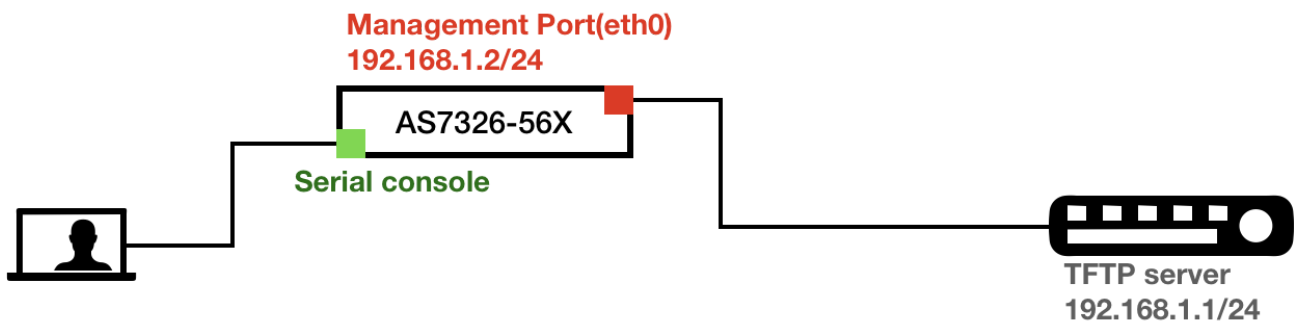
202012.2

Restriction:

- Suggest you should not install different branch SONiC image on switch, it may cause some exception issue.
- If your switch installed SONiC, you login to "ONIE :install OS" mode via grub, it will let the grub miss SONiC option.

Install SONiC from HTTP or TFTP via ONIE

Topology:



Procedure:

Step 1. Enter the ONIE install mode

Note: Switch shall automatically enter the ONIE install mode if there's no NOS installed yet.

GNU GRUB version 2.02

```
+-----+
|*ONIE: Install OS|
| ONIE: Rescue   |
| ONIE: Uninstall OS|
| ONIE: Update ONIE|
| ONIE: Embed ONIE|
| DIAG: Accton Diagnostic (accton_as7326_56x)|
|               |
|               |
|               |
|               |
|               |
+-----+
```

Use the ^ and v keys to select which entry is highlighted.
Press enter to boot the selected OS, `e` to edit the commands
before booting or `c` for a command-line.
The highlighted entry will be executed automatically in 2s.

Caution:

Since switch will automatically start the ONIE Service Discovery, this command can make user types easily.

This is not a necessary command. It won't affect installation no matter user execute it or not.

```
ONIE:/ # onie-discovery-stop  
discover: installer mode detected.  
Stopping: discover... done.
```

Step 3. Setup the ip address binding to switch management port

```
ONIE:/ # ifconfig eth0 192.168.1.2 netmask 255.255.255.0
```

Caution: Default ONIE will get ip from DHCP

Step 4. Install the image from remote URL via HTTP or TFTP

```
ONIE:/ # onie-nos-install tftp://192.168.1.1/Edgecore-SONiC_20211125_074752_ec202012_227.bin
```

If the installation is successful, the device will reboot automatically and boot-up with SONiC.

After finish the installation, user might need to check this article.

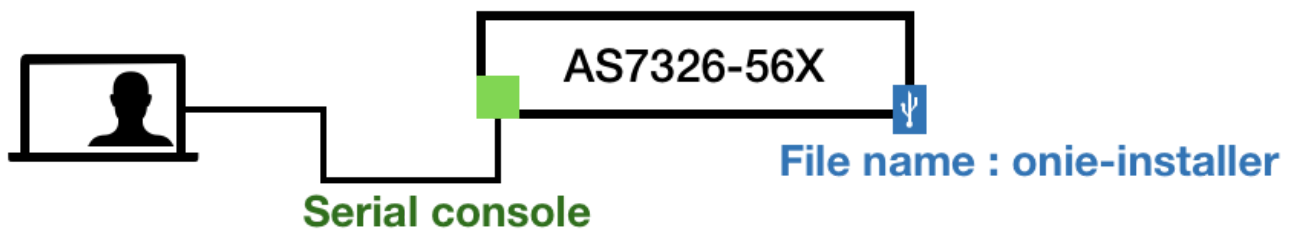
1. Username and Password
2. ecSONiC License Installation (License is no longer required from *SONiC.Edgecore-SONiC 201911.2 2020 Aug* version and later)

Install SONiC from USB via ONIE

Restriction:

- The ONIE don't support NTFS file system.

Topology:



Procedure:

Step 1. Copy Edgecore SONiC installer file to USB flash drive with a name "**onie-installer**".

Step 2. Reboot switch and plug in USB drive to switch

Step 3. Enter ONIE install mode

GNU GRUB version 2.02

```
+-----+
|*ONIE: Install OS|
| ONIE: Rescue    |
| ONIE: Uninstall OS|
| ONIE: Update ONIE|
| ONIE: Embed ONIE|
| DIAG: Accton Diagnostic (accton_as7326_56x)|
|                |
|                |
|                |
|                |
|                |
+-----+
```

Use the ^ and v keys to select which entry is highlighted.
Press enter to boot the selected OS, `e' to edit the commands
before booting or `c' for a command-line.
The highlighted entry will be executed automatically in 2s.

Step 4. Wait for ONIE to discover USB and start automatically uploading new firmware via USB drive.

```
ONIE: Starting ONIE Service Discovery
Info: Attempting file://dev/sdb1/onie-installer-x86_64-accton_as7326_56x-r0 ...
Info: Attempting file://dev/sdb1/onie-installer-x86_64-accton_as7326_56x-r0.bin ...
Info: Attempting file://dev/sdb1/onie-installer-x86_64-accton_as7326_56x ...
Info: Attempting file://dev/sdb1/onie-installer-x86_64-accton_as7326_56x.bin ...
Info: Attempting file://dev/sdb1/onie-installer-accton_as7326_56x ...
Info: Attempting file://dev/sdb1/onie-installer-accton_as7326_56x.bin ...
Info: Attempting file://dev/sdb1/onie-installer-x86_64-bcm ...
Info: Attempting file://dev/sdb1/onie-installer-x86_64-bcm.bin ...
Info: Attempting file://dev/sdb1/onie-installer-x86_64 ...
Info: Attempting file://dev/sdb1/onie-installer-x86_64.bin ...
```

```

Info: Attempting file://dev/sdb1/onie-installer ...
Info: Attempting file://dev/sdb1/onie-installer.bin ...
Info: Attempting file://dev/sdb/onie-installer-x86_64-accton_as7326_56x-r0 ...
Info: Attempting file://dev/sdb/onie-installer-x86_64-accton_as7326_56x-r0.bin ...
Info: Attempting file://dev/sdb/onie-installer-x86_64-accton_as7326_56x ...
Info: Attempting file://dev/sdb/onie-installer-x86_64-accton_as7326_56x.bin ...
Info: Attempting file://dev/sdb/onie-installer-accton_as7326_56x ...
Info: Attempting file://dev/sdb/onie-installer-accton_as7326_56x.bin ...
Info: Attempting file://dev/sdb/onie-installer-x86_64-bcm ...
Info: Attempting file://dev/sdb/onie-installer-x86_64-bcm.bin ...
Info: Attempting file://dev/sdb/onie-installer-x86_64 ...
Info: Attempting file://dev/sdb/onie-installer-x86_64.bin ...
Info: Attempting file://dev/sdb/onie-installer ...
Info: Attempting file://dev/sdb/onie-installer.bin ...
ONIE: Executing installer: file://dev/sdb1/onie-installer.bin
Verifying image checksum ... OK.
Preparing image archive ... OK.
Installing SONiC in ONIE
ONIE Installer: platform: x86_64-broadcom-r0
onie_platform: x86_64-accton_as7326_56x-r0

```

If the installation is successful, the device will reboot automatically and boot-up with SONiC.

After finish the installation, user might need to check this article.

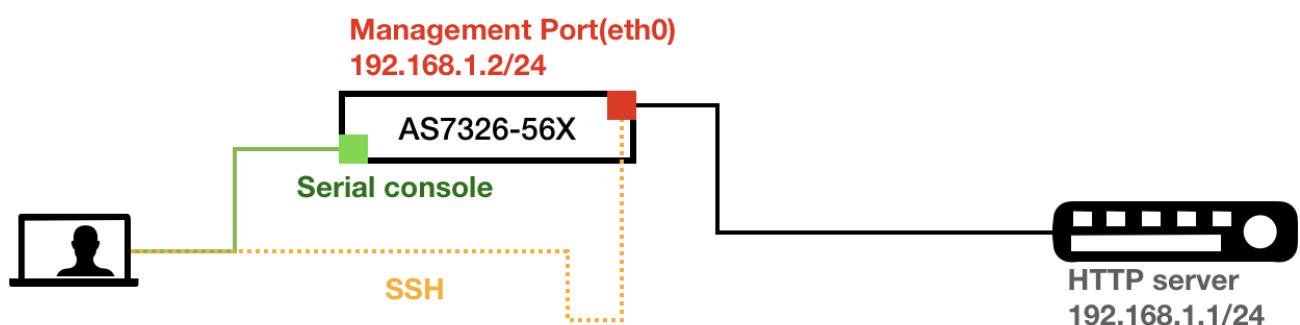
1. [Username and Password](#)
2. [ecSONiC License Installation \(License is no longer required from SONiC.Edgecore-SONiC 201911.2 2020 Aug version and later\)](#)

Edgecore SONiC Upgrade

Restriction:

If the switch have different branch image, please reset configuration in new image. (Detail refer the section of "Installation / Upgrade" in release note)

Topology:



Step 1. Check the image

```
admin@sonic:~$ sudo sonic_installer list
Warning: 'sonic_installer' command is deprecated and will be removed in the future
Please use 'sonic-installer' instead
Current: SONiC-OS-Edgecore-SONiC_20211125_074752_ec202012_227
Next: SONiC-OS-Edgecore-SONiC_20211125_074752_ec202012_227
Available:
SONiC-OS-Edgecore-SONiC_20211125_074752_ec202012_227
```

Step 2. Set the management IP (Refer to [Management and front port IPv4/IPv6 Address](#))

```
admin@sonic:~$ sudo config interface ip add eth0 192.168.1.2/24
```

Step 3. Upgrade image from remote HTTP server

```
admin@sonic:~$ sudo sonic_installer install http://192.168.1.1/Edgecore-SONiC_20210611_061351_ec202006_24
```

Caution:

- It could only use HTTP to upgrade.
- If the current switch configuration include "management vrf" and the http server connect to management port, please do update via "ip vrf exec mgmt" command as bellowing.

```
admin@sonic:~$ sudo ip vrf exec mgmt sonic_installer install http://192.168.1.3
```

Step 4. Check the images.

```
admin@sonic:~$ sudo sonic_installer list
Warning: 'sonic_installer' command is deprecated and will be removed in the future
Please use 'sonic-installer' instead
Current: SONiC-OS-Edgecore-SONiC_20211125_074752_ec202012_227
Next: SONiC-OS-Edgecore-SONiC_20210611_061351_ec202006_242
Available:
SONiC-OS-Edgecore-SONiC_20210611_061351_ec202006_242
SONiC-OS-Edgecore-SONiC_20211125_074752_ec202012_227
```

Note: The "Next" boot-up image shall be set with new image automatically.

Step 5. Reboot device

```
admin@sonic:~$ sudo reboot
```

Step 6. Check the images status

```
admin@sonic:~$ sudo sonic_installer list
Current: SONiC-OS-Edgecore-SONiC_20210611_061351_ec202006_242
Next: SONiC-OS-Edgecore-SONiC_20210611_061351_ec202006_242
Available:
SONiC-OS-Edgecore-SONiC_20210611_061351_ec202006_242
SONiC-OS-Edgecore-SONiC_20211125_074752_ec202012_227
```

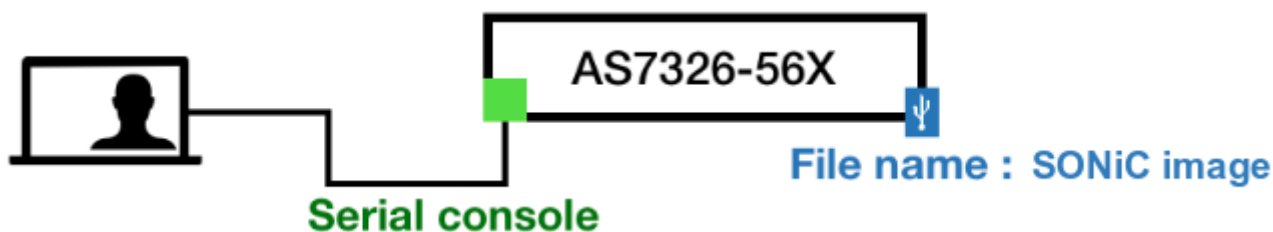
Edgecore SONiC Upgrade via USB

Restriction:

SONiC doesn't support the filesystem type 'NTFS'.

```
root@SWL:/mnt# mount /dev/sdb1 /mnt/test
mount: /mnt/test: unknown filesystem type 'ntfs'.
```

Topology:



Procedure:

Step 1: Confirm the position of the USB.

```
root@sonic:~# fdisk -l
Disk /dev/sda: 59.6 GiB, 64023257088 bytes, 125045424 sectors
Disk model: TS64GMTS400
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: gpt
Disk identifier: CADCD667-AE6B-4F9F-8969-A00D3ADBDBC0
```

Device	Start	End	Sectors	Size	Type
/dev/sda1	2048	526335	524288	256M	EFI System
/dev/sda2	526336	788479	262144	128M	ONIE boot
/dev/sda3	788480	1050623	262144	128M	Linux filesystem
/dev/sda4	1050624	68159487	67108864	32G	Linux filesystem

```
Disk /dev/loop0: 405 MiB, 424693760 bytes, 829480 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
```

```
Disk /dev/loop1: 4 GiB, 4294967296 bytes, 8388608 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
```

```
Disk /dev/sdb: 3.8 GiB, 4043309056 bytes, 7897088 sectors
Disk model: USB Flash Drive
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x94723316
```

Device	Boot	Start	End	Sectors	Size	Id	Type
/dev/sdb1		128	7897087	7896960	3.8G	c	W95 FAT32 (LBA)

Step 2: Create a folder and mount the USB to the folder that you create.

```
root@sonic:~# mkdir /mnt/test
root@sonic:~# mount /dev/sdb1 /mnt/test
root@sonic:~# cd /mnt/test
root@sonic:/mnt/test# ls
202012_279.bin
```

Step 3: Upgrade image from the USB

```
root@sonic:/mnt/test# sudo sonic_installer install /mnt/test/202012_279.bin
Warning: 'sonic_installer' command is deprecated and will be removed in the fut
Please use 'sonic-installer' instead
New image will be installed, continue? [y/N]: y
```

Step 4: Check the images.


```
root@sonic:/mnt/test# sudo sonic_installer list
Warning: 'sonic_installer' command is deprecated and will be removed in the future
Please use 'sonic-installer' instead
Current: SONiC-OS-Edgecore-SONiC_20220623_052151_ec202012_362
Next: SONiC-OS-Edgecore-SONiC_20220218_065711_ec202012_279
Available:
SONiC-OS-Edgecore-SONiC_20220218_065711_ec202012_279
SONiC-OS-Edgecore-SONiC_20220623_052151_ec202012_362
```

Step 5. Reboot device

```
admin@sonic:~$ sudo reboot
```

Specified the startup image

Procedure:

Step 1. Check the images status

```
admin@sonic:~$ sudo sonic_installer list
Current: SONiC-OS-Edgecore-SONiC_20210611_061351_ec202006_242
Next: SONiC-OS-Edgecore-SONiC_20210611_061351_ec202006_242
Available:
SONiC-OS-Edgecore-SONiC_20210611_061351_ec202006_242
SONiC-OS-Edgecore-SONiC_20211125_074752_ec202012_227
```

Caution:

- "Current" image, it mean the running image
- "Next" image, it mean the next boot up image.
- "Available" images, it mean installed SONiC images.

Step 2. Change the default boot-up image

```
admin@sonic:~$ sudo sonic_installer set_default SONiC-OS-Edgecore-SONiC_20211125_074752_ec202012_227
Command: grub-set-default --boot-directory=/host 1
```

Step 2. Check the image status

```
admin@sonic:~$ sudo sonic_installer list
Current: SONiC-OS-Edgecore-SONiC_20210611_061351_ec202006_242
```

```
Next: SONiC-OS-Edgecore-SONiC_20211125_074752_ec202012_227
```

Available:

```
SONiC-OS-Edgecore-SONiC_20210611_061351_ec202006_242
```

```
SONiC-OS-Edgecore-SONiC_20211125_074752_ec202012_227
```

Then the "Next" boot-up image changed.

Caution:

You can choose the boot-up image in GRUB menu directly, but this way won't change the default boot-up image.

Remove installed image

Procedure:

Step 1. Check the images status

```
admin@sonic:~$ sudo sonic-installer list
Current: SONiC-OS-Edgecore-SONiC_20211125_074752_ec202012_227
Next: SONiC-OS-Edgecore-SONiC_20211125_074752_ec202012_227
Available:
SONiC-OS-Edgecore-SONiC_20210611_061351_ec202006_242
SONiC-OS-Edgecore-SONiC_20211125_074752_ec202012_227
```

Step 2. Remove the image

```
admin@sonic:~$ sudo sonic-installer remove SONiC-OS-Edgecore-SONiC_20210611_061351_ec202006_242 -y
Updating GRUB...
Done
Removing image root filesystem...
Done
Command: grub-set-default --boot-directory=/host 0

Image removed
```

Step 3. Check the images status

```
admin@sonic:~$ sudo sonic-installer list
Current: SONiC-OS-Edgecore-SONiC_20211125_074752_ec202012_227
Next: SONiC-OS-Edgecore-SONiC_20211125_074752_ec202012_227
Available:
SONiC-OS-Edgecore-SONiC_20211125_074752_ec202012_227
```



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